MIL-D-46404 (Ord)
26 October 1960

MILITARY SPECIFICATION

DETONATOR, STAB, M13
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of stab detonator designated as Detonator, M13.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

7548102 - Box, Packing, Ammunition for Detonators
7548103 - Box, Set-up, Packing, Ammunition, Outer, for Boosters, Detonators and Primers. Assembly, Details, Packing and Marking.
7548104 - Box, Set-up, Packing, Ammunition, Inner, for Boosters, Detonators and Primers. Assembly, Details, Packing and Marking.
8797744 - Detonator, M13, Loading Assembly.

FSC 1375

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-6119351 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonators.-The detonators shall comply with all requirements specified on Drawing (dwg.) 8787744, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide.-The moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent when tested as specified in 4.3.2.

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3.4 Moisture content of primer mixture.-The moisture content of the primer mixture used in loading the detonators shall not exceed 0.30 percent when tested as specified in 4.3.3.

3.5 Functioning.

3.5.1 Sensitivity.-The detonators shall function on the first attempt and shall cause perforation of the lead disc to a diameter of not less than 1/4 inch when tested as specified in 4.3.5.1.

3.5.2 Waterproofness.-The detonators shall be waterproof as evidenced by normal functioning when tested as specified in 4.3.5.2.

3.6 Workmanship.-All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description

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will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. Except in the case of the 4.3.5.1 the contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In the case of the sensitivity test, the weight of the ball and height of drop shall not differ from the values specified in 4.3.5.1 and on the applicable test equipment drawing. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist

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of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instnace will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. (See Specification MIL-G-2550)

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of detonators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot

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number. Each lot shall contain:

a. Parts bearing the same interfix lot number from one manufacturer.

b. Lead azide from not more than one lot.

c. Primer mixture from one batch or blending of batches that results in homogeneity of all primer mixture used in loading the entire detonator lot.

d. Tetryl from not more than one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, prior to loading (see dwg. 8797744).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Coating compound or lacquer on exposed surface of disc	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.2 Assembly (see dwg. 8797744).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Length	Gage
102.	Diameter, maximum	Gage
103.	Color of lacquer incorrect (either ends)	Visual

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104.	Disc loose	Manual
105.	Not crimped full 360 degrees	Visual
106.	Disc punctured or torn	Visual
107.	Cup damaged	Visual

Minor: AQL 0.40 percent

201.	Evidence of poor workmanship (see 3.6).	Visual

4.2.2.3 Box, set-up, packing, ammunition, inner (see dwg. 7548104).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Box damaged exposing contents	Visual
102.	Label missing or incorrect	Visual
Minor:	None defined	

4.2.2.4 Box, set-up, packing, ammunition, outer (see dwg. 7548103).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Barrier bag damaged	Visual
102.	Label missing or incorrect	Visual
Minor:	None defined	

4.2.2.5 Box, packing, ammunition, for detonators (see dwg. 7548102).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Strapping missing or loose	Visual-Manual
102.	Box damaged	Visual
103.	Nails missing or improperly driven	Visual
104.	Marking incomplete or misleading	Visual
Minor:	None defined	

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4.2.3 Testing.

4.2.3.1 Binder-lubricant content of tetryl (see dwg. 8797744) - Major defect.-The binder-lubricant content of the tetryl shall be determined for each lot at the time of loading. A sample of 5 grams shall be selected and tested as specified in 4.3.1. If the binder-lubricant content is in excess of the percentage specified, and loading has not begun, the lot of tetryl shall be rejected until proper corrective action has been accomplished as verified by repeating the binder-lubricant content test specified in 4.3.1. If loading has begun, action on the unloaded tetryl shall be as specified above and all detonators loaded with the questionable tetryl shall be rejected.

4.2.3.2 Moisture content of lead azide (see 3.3) - Major - defect.-The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.3.2. If the moisture content is in excess of the percentage specified and if loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.2. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.2.3.3 Primer mixture.

4.2.3.3.1 Moisture content (see 3.4) - Major defect.-The moisture content of the primer mixture shall be determined for each lot at the time of loading. A sample of 0.7 gram shall be selected and tested as specified in 4.3.3. If the moisture content is in excess of the percentage specified and if loading has not begun, the lot of primer mixture shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.3. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.2.3.3.2 Composition (see dwg. 8797744) - Major defect.-The composition of the primer mixture shall be determined for each lot at the time of loading. A sample of 0.5 gram shall be selected and tested as specified in 4.3.4. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing, and if loading has not begun, the lot of primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable mixture shall be rejected.

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4.2.3.4 Functioning.

4.2.3.4.1 Sensitivity (see 3.5.1) - Major defect.-A sample of complete detonators shall be selected in accordance with Standard MIL-STD-105 using code letter P and an AQL of 0.15 percent. The sample detonators shall be tested as specified in 4.3.5.1, using test equipment specified in 4.2.4. Rejection shall be in accordance with Standard MIL-STD-105.

4.2.3.4.2 Waterproofness (see 3.5.2) - Major defect.-A sample of 25 detonators shall be selected from each lot and subjected to the test specified in 4.3.5.2, using test equipment specified in 4.2.4. If one or more detonators fail to function satisfactorily, as evidenced by performing the test specified in 4.3.5.1, the lot shall be rejected.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Lists No. 6119351, or approved equivalents thereto. (See 4.1.1) When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Determination of binder-lubricant content of tetryl.-An accurately weighed portion of 5 grams shall be transferred to a tared, medium porosity, sintered glass filtering crucible. The sample shall be extracted with four 10 milliliter (ml.) portions of acetone maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible washed down with each 10 ml. portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees Centigrade (°C.) to 150°C. for 1/2 hour, cooled in a desiccator and weighed. The

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above extraction shall be repeated, drying and weighing with individual 10 ml. portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of residue in the crucible shall be calculated as the percentage of total binder-lubricant.

4.3.2 Determination of moisture content of lead azide.-A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrafluoroethylene 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams of the sample shall be transferred to the moisture dish and the dish and contents shall be transferred to an oven and dried at 95°C. ± 2°C. for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrafluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated as percentage of moisture in the lead azide to determine compliance with 3.3 as follows:

$$\text{percentage of moisture} = \frac{(A-B)}{A} \times 100$$

where:

A = the weight of the sample before drying.
B = the weight of the sample after drying.

4.3.3 Moisture content of primer mixture.-The specified sample shall be accurately weighed to the nearest milligram and dried to a constant weight at 60°C. ± 5°C. or in a desiccator containing calcium chloride. The loss in weight shall be calculated as the percentage of moisture.

4.3.4 Composition of primer mixture.

4.3.4.1 Potassium chlorate.-Approximately 1 gram of the primer mixture shall be accurately weighed and transferred to a tared crucible of approximately 15 ml. capacity. Three ml. of lead azide-saturated distilled water shall be added at a temperature of 25°C. ± 2°C., agitated for exactly 1 minute, and if necessary, any lumps shall be broken up with a glass rod. Suction shall be applied. This procedure shall be repeated five times making a total of six 3 ml. extractions. The sample shall be rinsed in the filtering crucible with a few ml. of alcohol and then with a few ml. of ether, dried for approximately 15 minutes at 95°C. ± 2°C., cooled in a desiccator and weighed. The loss in weight shall be calculated

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as the percentage of potassium chlorate.

4.3.4.2 Lead azide.-The residue remaining from the potassium chlorate determination shall be extracted with 5 ml. portions of saturated ammonium acetate solution at about 25°C. (hot saturated ammonium acetate solution must not be used as it dissolves antimony sulfide to some extent) each 5 ml. portion being agitated for about 30 seconds. Extractions shall be continued with 5 ml. portions until the washing no longer give a yellow precipitate with the addition of a few drops of potassium dichromate to the solution. The use of more than 60 ml. (12 washings) of ammonium acetate solution should be avoided. The residue shall be washed with distilled water, rinsed with a few ml. of alcohol and then with a few ml. of ether, dried for approximately 15 minutes at 100°C. ± 5°C., cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of lead azide.

4.3.4.3 Antimony sulfide.-The residue remaining from the lead azide determination shall be extracted with concentrated hydrochloric acid until free from antimony sulfide. (This extraction can be made without removing the sample from the filtering crucible) Suction shall be applied and most of the antimony sulfide extracted with cool concentrated hydrochloric acid, then rinsed thoroughly, ignited to separate fulfur and organic matter, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of antimony sulfide.

4.3.4.4 Carborundum.-The original weight of the ignited filtering crucible shall be subtracted from the weight of the crucible and residue after the antimony sulfide determination and the difference calculated as the percentage of carborundum.

4.3.5 Functioning.

4.3.5.1 Sensitivity.-The detonator shall be assembled to the applicable test fixture specified in 4.2.4, so that the firing pin will strike the sensitive end of the detonator. A steel ball 1 3/16 inch in diameter and weighing 3.95 ounces maximum, shall be dropped on the firing pin, to function the detonator, from a height of 12 inches maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

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4.3.5.2 Waterproofness.-The detonators shall be immersed under 2 inches of water for not less than 48 hours. Within one hour after removal from the water the detonators shall be subjected to the sensitivity test specified in 4.3.5.1. Any detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Levels B (see 6.2) and C.-The detonator shall be preserved, marked and packaged in accordance with dwg. 7548103 and dwg. 7548104.

5.2 Packing and marking.

5.2.1 Level C.-The detonators shall be packed and marked in accordance with dwg. 7548102.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Level B packaging.-In the event the expected storage time is in excess of 180 days the barrier bag shall be used as specified on the applicable drawing. Expected storage time shall be determined by the procuring activity.

6.3 Check test for possible deterioration.-The following test is designed to prevent the loading of detonators which have deteriorated into a major ammunition item. The test should be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor reimbused for any expense incurred in the performance of the check test for possible deterioration.

6.3.1 Sensitivity.-A sample of detonators shall be selected in accordance with Standard MIL-STD-105, using code letter P and an AQL of 0.25 percent. The sample detonators shall be tested as specified in 4.3.5.1, using test equipment specified in 4.2.4. Rejection shall be in accordance with Standard MIL-STD-105.

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